Maxima & Minima: Dynamic Motion & Optimization

Differential Calculus for Engineering // Visual Storytelling in Motion

KEY CONCEPTUAL & MATHEMATICAL TAKEAWAYS

- Derivatives as velocity and direction detectors along physical curves
- Fermat's Theorem: Horizontal tangents where f'(x) = 0
- Core visual model: Cubic Landscape f(x) = x³ - 3x
- Primary source: MATHalino Differential Calculus Review

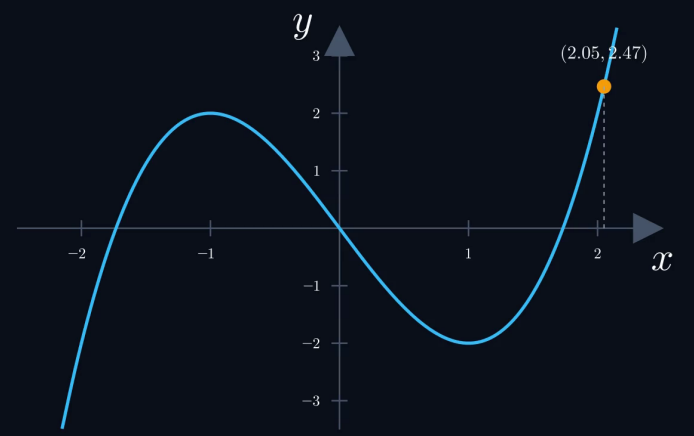
LECTURER TELEPROMPTER & TEACHING HINTS

Welcome students. Introduce the duality between geometric curvature and engineering extremum principles.

The Cubic Function Landscape

Tracing vertical displacement and turning behavior along f(x) = x³ - 3x

$f(x) = x^3 - 3x$



APPLE PENCIL LIVE WORKSPACE *[GoodNotes / Notability / Keynote inking area]*

Lecture Trajectory: From Kinetic Intuition to Optimization

The 4 Pillars of Differential Calculus Extrema

KEY CONCEPTUAL & MATHEMATICAL TAKEAWAYS

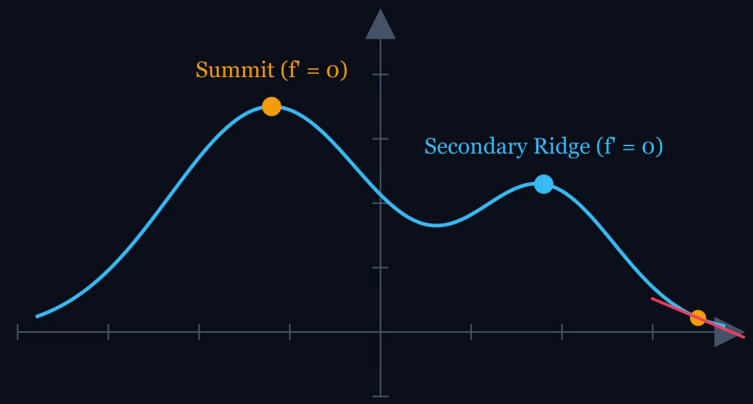
- 01: The Geometry of Turning (Fermat's Theorem & critical numbers)
- 02: Classification Engines (1st Derivative Sign Chart vs 2nd Derivative Concavity)
- 03: Boundary Constraints (Closed Interval Method on [a, b])
- 04: Applied Engineering Optimization (Mathalino Real-World Case Studies)

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Walk through the four learning modules. Emphasize that solving engineering problems requires both finding critical points and classifying them.

The Geometry of Nature: Mountain Terrains

Where does physical motion pause? Where are the peaks?



APPLE PENCIL LIVE WORKSPACE *[GoodNotes / Notability / Keynote inking area]*

Intuitive Meaning of Extrema: Peaks, Valleys & Equilibrium

Mechanical Energy Wells, Circuit Matching & Thermal Extremes

KEY CONCEPTUAL & MATHEMATICAL TAKEAWAYS

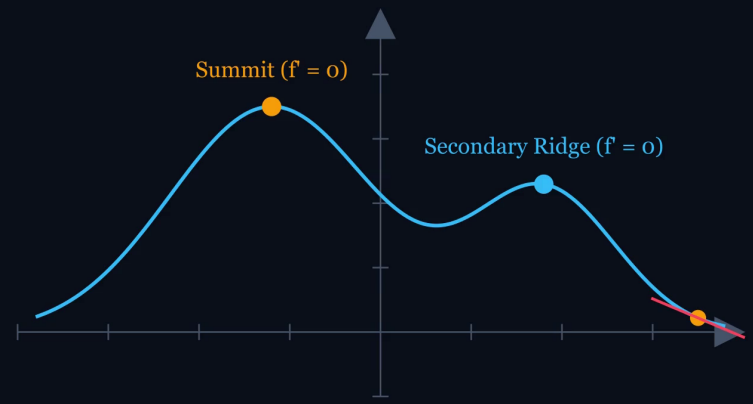
- Mechanical systems: Stable equilibrium at potential energy minima
- Unstable equilibrium at potential energy peaks (dU/dx = 0)
- Electrical systems: Maximum power transfer at matched impedance
- Stationary states occur where net restoring force vanishes

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Ask students: 'If a derivative is zero, does the system definitely have a peak or valley?' Elicit guesses about inflection points.

The Geometry of Nature: Mountain Terrains

Where does physical motion pause? Where are the peaks?



APPLE PENCIL LIVE WORKSPACE *[GoodNotes / Notability / Keynote inking area]*

Local vs. Absolute Extrema: Global Domain Hazards

Relative Neighborhood Extremes vs. System Operating Envelopes

KEY CONCEPTUAL & MATHEMATICAL TAKEAWAYS

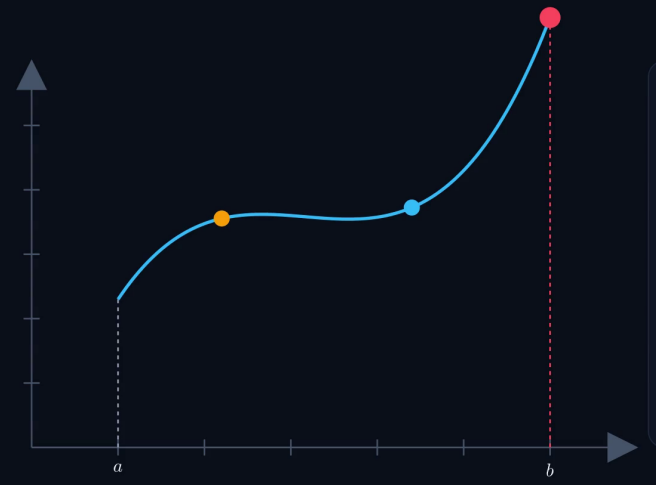
- Local extremum: Peak or valley relative to immediate neighborhood (x - δ, x + δ)
- Absolute extremum: Highest or lowest point across entire operating domain [a, b]
- Boundary endpoints can exceed all interior local peaks
- Unbounded domains may have local peaks but no absolute maximum

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Highlight how confusing local and absolute extrema causes structural and circuit over-design failures.

Local vs. Absolute (Global) Extrema

Neighborhood champions vs. the entire physical domain [a, b]



Extrema Taxonomy

LOCAL MAXIMUM (Interior)
Tallest in immediate open neighborhood (c-δ, c+δ)

LOCAL MINIMUM (Interior)
Lowest in immediate open neighborhood

ABSOLUTE MAXIMUM (at boundary b)
Highest point over ENTIRE domain [a, b]!
Notice: Occurs where f'(x) ≠ 0!

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The Canonical Cubic Landscape: f(x) = x³ - 3x

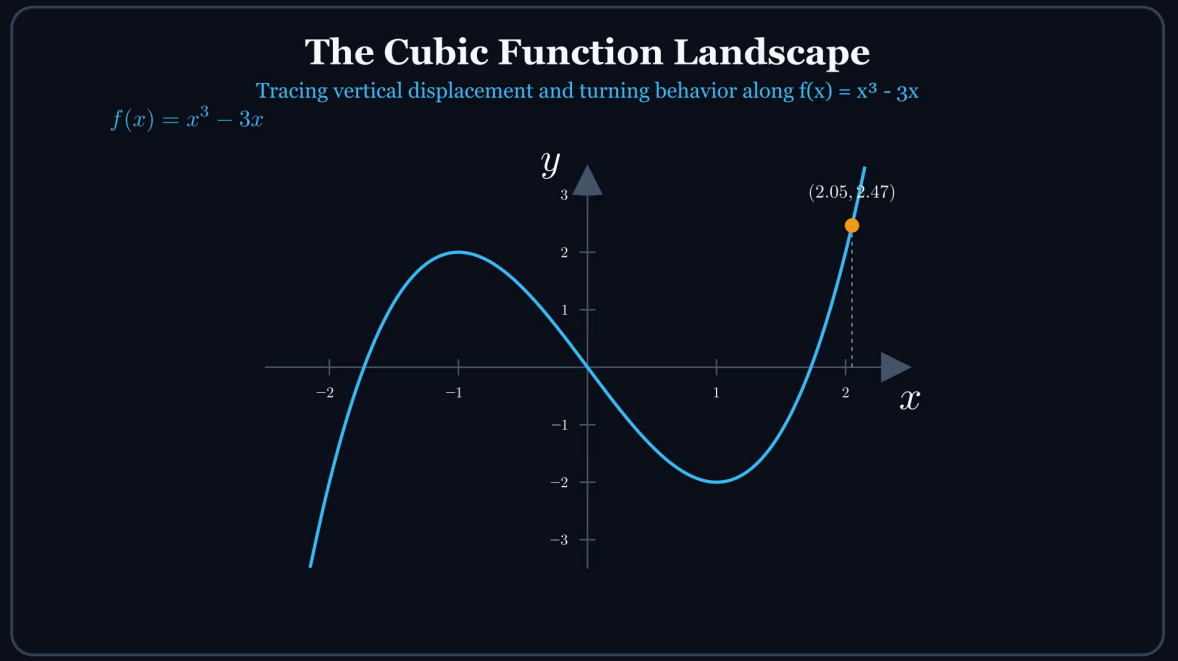
Our Reference Model for Visualizing Calculus Operations

KEY CONCEPTUAL & MATHEMATICAL TAKEAWAYS

- Core function: f(x) = x³ - 3x
- Polynomial continuity ensures smooth differentiability everywhere
- Local maximum located at x = -1, f(-1) = 2
- Local minimum located at x = +1, f(+1) = -2
- Inflection point and rotational symmetry center at the origin (0, 0)

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Trace the cubic curve on the iPad screen. Have students notice the odd-function symmetry f(-x) = -f(x).



APPLE PENCIL LIVE WORKSPACE *[GoodNotes / Notability / Keynote inking area]*

Dynamic Tangent Telemetry: The Derivative in Motion

Tracking Slope Vector and Tangent Line Across the Curve

KEY CONCEPTUAL & MATHEMATICAL TAKEAWAYS

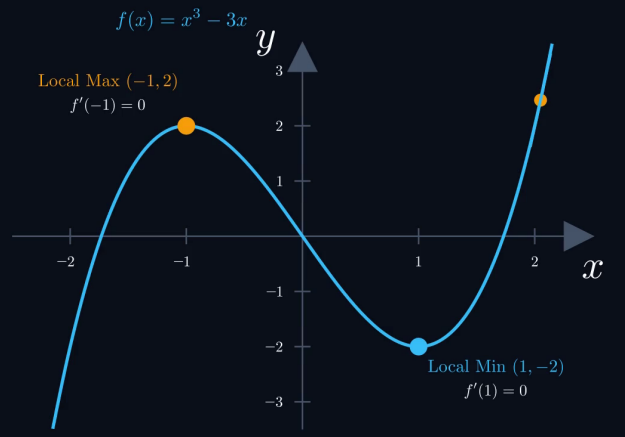
- Tangent slope m = f'(x) represents instantaneous rate of change
- m > 0 (Green): Function is strictly increasing
- m < 0 (Coral): Function is strictly decreasing
- m = 0 (Cyan): Tangent is horizontal (Stationary point)

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Point out how the tangent line rocks from positive slope, flattens to horizontal at x = -1, turns negative, and flattens again at x = +1.

Dynamic Tangent & Instantaneous Slope

Vanishing rate of change at turning points: f'(x) = 0



Real-Time Telemetry

$x = 2.05$

$f'(x) = +9.61$

Ascending (f' > 0)

APPLE PENCIL LIVE WORKSPACE *[GoodNotes / Notability / Keynote inking area]*

Critical Numbers & Singularities: Where f'(x) DNE

Sharp Corners, Cusps, and Vertical Tangents

KEY CONCEPTUAL & MATHEMATICAL TAKEAWAYS

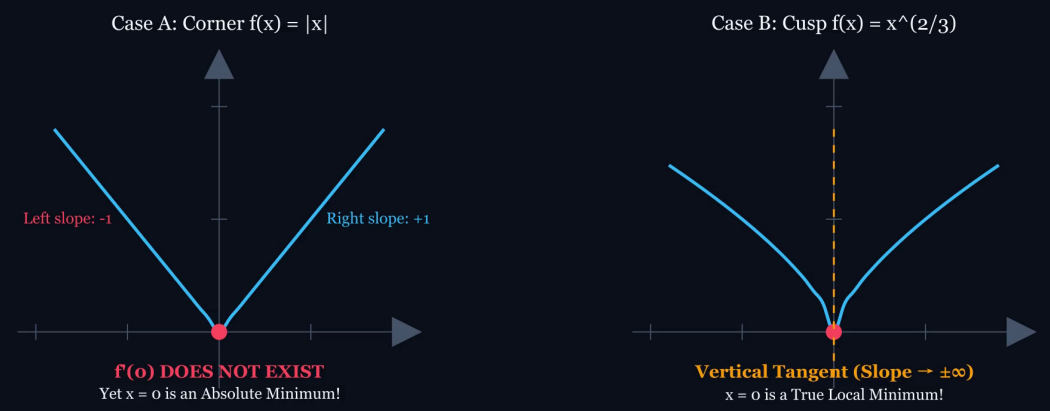
- Definition: Critical numbers occur where f'(c) = 0 OR f'(c) does not exist
- Corner: Left-hand derivative ≠ Right-hand derivative (e.g. |x|)
- Cusp: Infinite derivatives of opposite signs on either side (e.g. x^(2/3))
- Singularities can be valid extrema despite zero horizontal slope

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Remind students: Calculus is not just setting derivative to zero! Never forget points where the derivative fails to exist.

Singular Points: When f'(x) Fails to Exist

Corners and cusps harbor crucial physical extrema



APPLE PENCIL LIVE WORKSPACE *[GoodNotes / Notability / Keynote inking area]*

The Critical Trap: Horizontal Tangents Without Extrema

Stationary Inflection Terraces: f(x) = x³ at the Origin

KEY CONCEPTUAL & MATHEMATICAL TAKEAWAYS

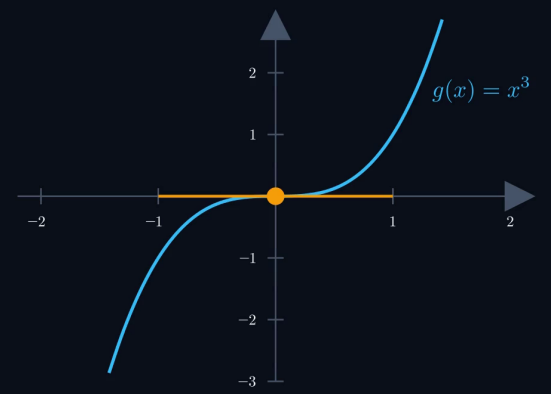
- Crucial Caveat: f'(c) = 0 is NECESSARY for smooth extrema, NOT SUFFICIENT
- Counterexample: f(x) = x³ has f'(0) = 0, but is neither max nor min
- The curve pauses horizontally, then continues climbing (Terrace)
- Classification test is strictly mandatory before declaring an extremum

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Stress this point heavily! Many engineering students assume f'(x) = 0 guarantees a maximum. Highlight the inflection terrace.

Inflection Terrace Counterexample

g'(0) = 0 without an extremum: The stationary inflection trap



Critical Analysis of g(x) = x³

$g'(x) = 3x^2$

$g'(0) = 3(0)^2 = 0$ (Horizontal)

Examine neighborhood slopes:

$x < 0 \implies g'(x) = 3x^2 > 0$ (↗)

$x > 0 \implies g'(x) = 3x^2 > 0$ (↗)

NO SIGN CHANGE across x = 0!

Origin is a STATIONARY INFLECTION point.

APPLE PENCIL LIVE WORKSPACE *[GoodNotes / Notability / Keynote inking area]*

Algebraic Morph: Factoring f'(x) = 3(x + 1)(x - 1)

Continuous Algebraic Morphing from Power Rule to Critical Numbers

KEY CONCEPTUAL & MATHEMATICAL TAKEAWAYS

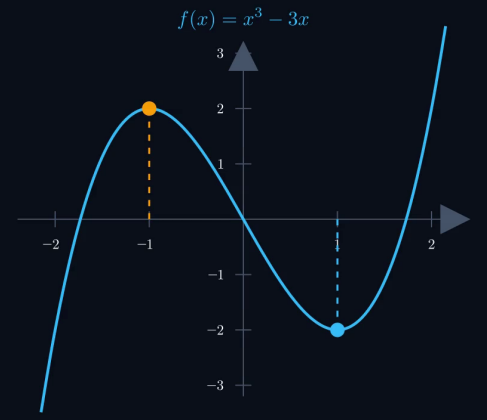
- Differentiating: d/dx [x³ - 3x] = 3x² - 3
- Factoring constant: 3(x² - 1)
- Difference of squares: 3(x + 1)(x - 1) = 0
- Exact critical numbers: x = -1 and x = +1

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Step through the factoring algebra. Emphasize how factoring partitions the real number line into test intervals.

Algebraic Factoring & Critical Roots

Factoring f'(x) to isolate critical numbers x = -1 and x = +1



Algebraic Factorization

$$f'(x) = 3(x-1)(x+1)$$

$$f'(x) = 0 \implies x = -1 \text{ or } x = +1$$

Zeros of f'(x) align with horizontal tangents:

$x_1 = -1$ (Peak), $x_2 = +1$ (Trough)

APPLE PENCIL LIVE WORKSPACE *[GoodNotes / Notability / Keynote inking area]*

The First Derivative Test: Sign Chart Engine

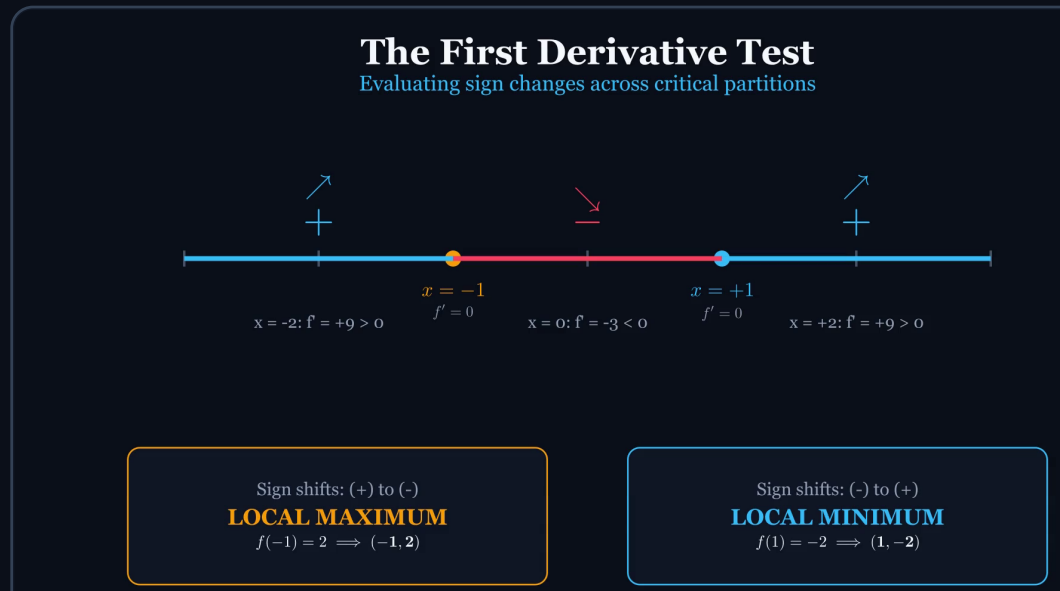
Testing Interval Slopes Across Partitioned Real Line

KEY CONCEPTUAL & MATHEMATICAL TAKEAWAYS

- Interval (-∞, -1): f'(-2) = 3(-1)(-3) = +9 > 0 (Increasing)
- Interval (-1, +1): f'(0) = 3(1)(-1) = -3 < 0 (Decreasing)
- Interval (+1, +∞): f'(+2) = 3(3)(1) = +9 > 0 (Increasing)
- Sign transition (+) to (-) indicates LOCAL MAXIMUM at x = -1
- Sign transition (-) to (+) indicates LOCAL MINIMUM at x = +1

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Guide students to fill in the sign chart on their iPads. Stress that sign changes reveal the shape of the function.



APPLE PENCIL LIVE WORKSPACE *[GoodNotes / Notability / Keynote inking area]*

Exact Coordinate Verification & Extremum Reveal

Substituting Critical Numbers into Original Objective Function

KEY CONCEPTUAL & MATHEMATICAL TAKEAWAYS

- Always evaluate y-coordinates in ORIGINAL f(x), never in f'(x)!
- At x = -1: f(-1) = (-1)³ - 3(-1) = -1 + 3 = +2 □ Local Max (-1, 2)
- At x = +1: f(+1) = (+1)³ - 3(+1) = 1 - 3 = -2 □ Local Min (1, -2)
- Extremum coordinates validated by Manim telemetry coordinate brackets

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Common student blunder: plugging critical x back into f'(x) and getting 0. Emphasize using original f(x).

Extrema Coordinates & Verification

Evaluating original function f(x) to anchor extrema heights



Exact Altitude Evaluation

1. Local Maximum Coordinate:

$f(-1) = (-1)^3 - 3(-1)$

$= -1 + 3 = +2$

$\Longrightarrow$ Point $(-1, 2)$

2. Local Minimum Coordinate:

$f(1) = (1)^3 - 3(1)$

$= 1 - 3 = -2$

$\Longrightarrow$ Point $(1, -2)$

□□ **APPLE PENCIL LIVE WORKSPACE** *[GoodNotes / Notability / Keynote inking area]*

Second Derivative Test: Curvature & Concavity

Differential Bending f''(x) = 6x and the Tangent Bowl Intuition

KEY CONCEPTUAL & MATHEMATICAL TAKEAWAYS

- Second derivative: f''(x) = d/dx [3x² - 3] = 6x
- At x = -1: f''(-1) = -6 < 0 □ Concave Down ∩ (Local Maximum)
- At x = +1: f''(+1) = +6 > 0 □ Concave Up □ (Local Minimum)
- If f''(c) = 0, the test is INCONCLUSIVE (Must use First Derivative Test)

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Use the bowl mnemonic: Concave up holds water (smiling, minimum), concave down spills water (frowning, maximum).

Second Derivative & Concavity Bending

Bending rate of change f''(x) = 6x verifies extrema

$$f''(x) = 6x$$



CONCAVE DOWN (∩)

$f''(-1) = 6(-1) = -6 < 0$

Slopes decreasing: Traps a

LOCAL MAXIMUM



CONCAVE UP (∪)

$f''(1) = 6(1) = +6 > 0$

Slopes increasing: Holds a

LOCAL MINIMUM

□□ **APPLE PENCIL LIVE WORKSPACE** *[GoodNotes / Notability / Keynote inking area]*

1st vs. 2nd Derivative Test Comparison Matrix

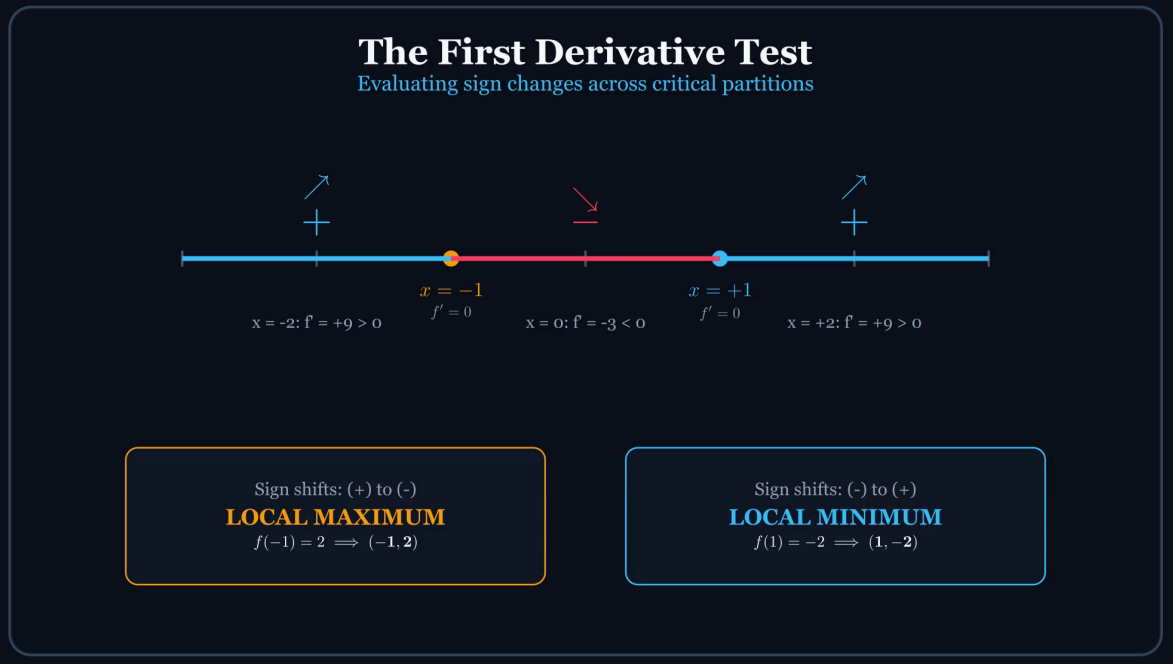
Decision Architecture: When to Use Which Classification Engine

KEY CONCEPTUAL & MATHEMATICAL TAKEAWAYS

- 1st Derivative Test: Works on ALL functions, handles corners and cusps
- Requires testing points in intervals between critical numbers
- 2nd Derivative Test: Faster when f''(x) is easy to compute
- Fails when f''(c) = 0 or f''(c) DNE; only applies to smooth horizontal tangents

LECTURER TELEPROMPTER & TEACHING HINTS

Present the engineering trade-off: 2nd derivative test is computationally cheaper for polynomials, but 1st test is bulletproof.



APPLE PENCIL LIVE WORKSPACE *[GoodNotes / Notability / Keynote inking area]*

The Closed Interval Method (EVT)

Extreme Value Theorem on [a, b]: Never Ignore the Endpoints!

KEY CONCEPTUAL & MATHEMATICAL TAKEAWAYS

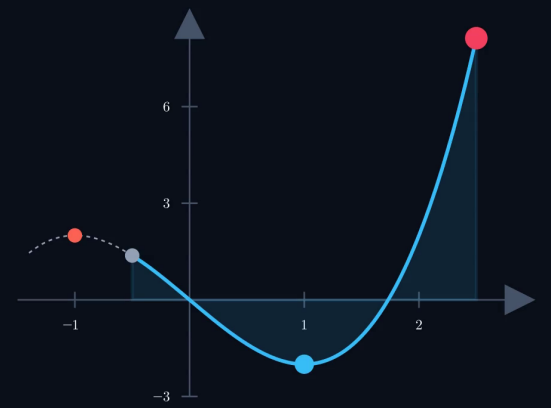
- Extreme Value Theorem: Continuous f(x) on closed [a, b] MUST attain absolute extrema
- 3-Step Protocol: 1. Evaluate f(c) at all interior critical points
- 2. Evaluate f(a) and f(b) at the boundary endpoints
- 3. The largest value is the ABSOLUTE MAX; the smallest is the ABSOLUTE MIN

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Demonstrate on the cubic on [-2, 2.5]. Show how the endpoint f(2.5) = 8.125 dwarfs the local peak f(-1) = 2.

The Closed Interval Method: f(x) = x³ - 3x on [-0.5, 2.5]

Testing interior critical candidates against domain boundary endpoints



Candidate Evaluation Matrix

1. Filter Critical Points:

$x = -1 \notin [-0.5, 2.5] \implies$ DISCARDED

$x = +1 \in (-0.5, 2.5) \implies$ VALID INTERIOR

2. Evaluate All Candidates:

$f(1) = -2.000 \implies$ ABSOLUTE MINIMUM

$f(-0.5) = 1.375$ (Left Endpoint)

$f(2.5) = 8.125 \implies$ ABSOLUTE MAXIMUM

Takeaway: True peak occurs at boundary!

APPLE PENCIL LIVE WORKSPACE *[GoodNotes / Notability / Keynote inking area]*

Engineering Application: Maximum Power Transfer

Optimizing Electrical Circuit Power Across Variable Load Resistance

KEY CONCEPTUAL & MATHEMATICAL TAKEAWAYS

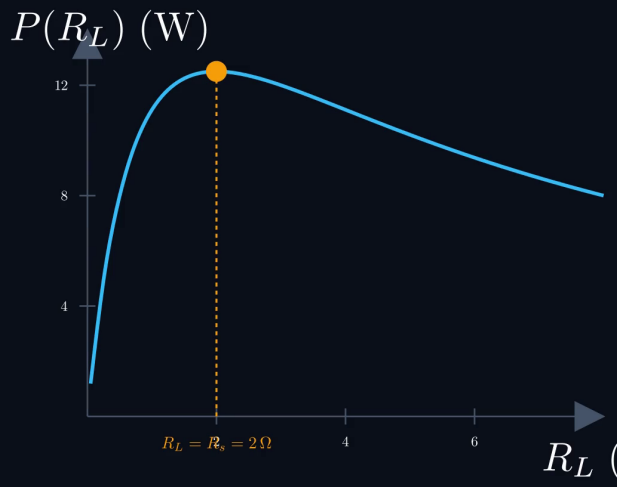
- Power delivered to load: P(R_L) = V_th² * R_L / (R_th + R_L)²
- Differentiating with Quotient Rule: dP/dR_L = V_th² (R_th - R_L) / (R_th + R_L)³
- Setting dP/dR_L = 0 yields impedance matching condition: R_L = R_th
- Maximum transferable power: P_max = V_th² / (4 R_th)

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Connect pure mathematics to electrical engineering. Ask: 'Why don't power companies match generator impedance on the power grid?'

Engineering Optimization: Maximum Power Transfer

Impedance matching in electrical, RF, and renewable energy circuits



Calculus of Impedance Matching

$$P(R_L) = \frac{V_s^2 R_L}{(R_s + R_L)^2}$$

Apply Quotient Rule:

$$\frac{dP}{dR_L} = V_s^2 \frac{R_s - R_L}{(R_s + R_L)^3}$$

Stationary Condition dP/dRL = 0:

$$R_s - R_L = 0 \implies \mathbf{R_L = R_s}$$

$$P_{\max} = \frac{V_s^2}{4R_s} = \mathbf{12.5\,W}$$

Governs audio, RF, & solar MPPT tracking!

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Mathalino Problem 01: Number Optimization

Find the number that exceeds its square by the maximum possible amount

KEY CONCEPTUAL & MATHEMATICAL TAKEAWAYS

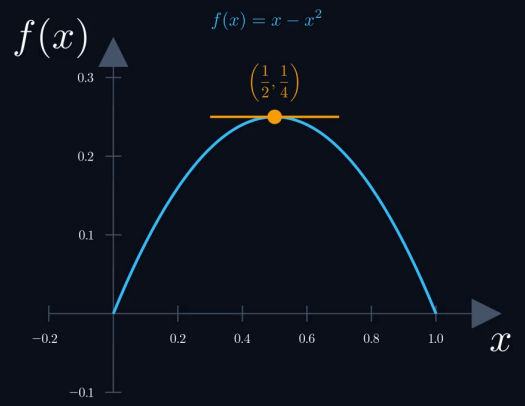
- Objective function: f(x) = x - x²
- Feasible domain: 0 < x < 1 (where x > x²)
- Differentiating: f'(x) = 1 - 2x = 0 □ x = 1/2
- Second derivative: f''(x) = -2 < 0 □ Confirmed absolute maximum
- Maximum excess: f(1/2) = 1/2 - 1/4 = 1/4

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Source: MATHalino Number Problems in Maxima and Minima. Emphasize setting up the single-variable function clearly.

Optimization: Number Exceeding its Square

Finding the number that exceeds its square by the maximum amount



Mathematical Modeling Workflow

1. Define: x = number, x^2 = square
2. Objective: $f(x) = x - x^2$
3. Differentiate: $f'(x) = 1 - 2x = 0$
4. Critical Number: $2x = 1 \implies x = \frac{1}{2}$
5. Maximum Excess: $f\left(\frac{1}{2}\right) = \frac{1}{2} - \frac{1}{4} = \frac{1}{4}$
6. Concavity: $f''(x) = -2 < 0 \implies$ **Strict Maximum**

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Mathalino Problem 02: Riverbank Rectangular Lot

Fencing 1,200 ft on Three Sides Along a Straight River for Maximum Area

KEY CONCEPTUAL & MATHEMATICAL TAKEAWAYS

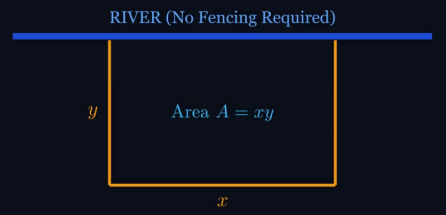
- Perimeter constraint (river requires no fencing): 2x + y = 1200 □ y = 1200 - 2x
- Objective Area function: A(x) = x(1200 - 2x) = 1200x - 2x²
- Differentiating: A'(x) = 1200 - 4x = 0 □ x = 300 ft
- Optimum dimensions: Width x = 300 ft, Length y = 600 ft
- Maximum enclosed area: A_max = 300 * 600 = 180,000 sq ft

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Source: MATHalino Rectangular Lot Problems. Show that the length is exactly twice the width for optimal 3-sided fencing.

Optimization: Riverbank Fencing Design

Minimum perimeter fence for fixed area A along a straight river



Geometric Constraint Reduction

1. Constraint: $A = xy \implies x = \frac{A}{y}$
2. Objective (Fence): $P = x + 2y$
3. Single Variable: $P(y) = \frac{A}{y} + 2y$
4. Differentiate: $\frac{dP}{dy} = -\frac{A}{y^2} + 2 = 0$
5. Solve: $2y^2 = A = xy \implies y = \frac{1}{2}x$
6. Optimum Shape: Width $\mathbf{y} = \frac{1}{2} \times$ Length $\mathbf{x}$
7. Minimum Check: $\frac{d^2P}{dy^2} = \frac{2A}{y^3} > 0 \implies$ **Minimum**

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Mathalino Problem 03: Open-Top Box from Cardboard

Cutting Square Corners from Sheet to Maximize Folded Box Volume

KEY CONCEPTUAL & MATHEMATICAL TAKEAWAYS

- Sheet dimensions: L0 = 24 in, W0 = 15 in; corner cutout = x
- Folded dimensions: Length = 24 - 2x, Width = 15 - 2x, Height = x
- Volume: V(x) = x(24 - 2x)(15 - 2x) = 4x³ - 78x² + 360x
- Derivative: V'(x) = 12x² - 156x + 360 = 12(x - 3)(x - 10) = 0
- Feasible domain x □ (0, 7.5) □ Critical point x = 3 in (x = 10 is impossible)

LECTURER TELEPROMPTER & TEACHING HINTS

Source: MATHalino Box Open at the Top. Highlight discarding x = 10 because a 15-inch sheet cannot have two 10-inch cuts.

Optimization: Open-Top Box Volume

Cardboard 15" × 24": Cut corner squares x to maximize Volume



Volume Objective & Critical Points

$$V(x) = 4x^3 - 78x^2 + 360x$$

$$V'(x) = 12x^2 - 156x + 360 = 0$$

$$x^2 - 13x + 30 = 0 \implies (x-10)(x-3) = 0$$

Feasible Domain: $0 < x < \frac{15}{2} = 7.5''$

$x = 10'' \implies$ **Impossible** $(15 - 20 < 0)$

Optimal Depth $: x = 3'' \in (0, 7.5)$

$$V_{\max} = (24-6)(15-6)(3) = \mathbf{486}\ \text{in}^3$$

□□ **APPLE PENCIL LIVE WORKSPACE** *[GoodNotes / Notability / Keynote inking area]*

Mathalino Problem 04: Tour Bus Maximum Revenue

80-Seat Tour Bus Pricing Elasticity & Revenue Maximization

KEY CONCEPTUAL & MATHEMATICAL TAKEAWAYS

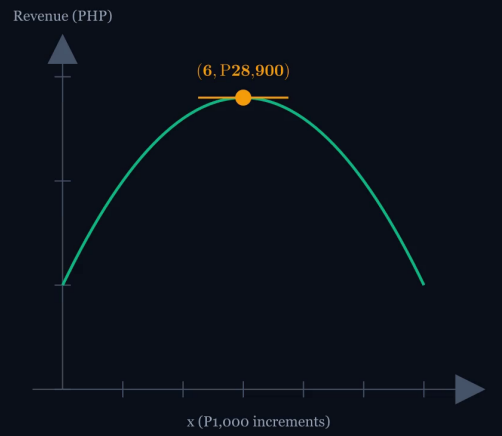
- Base fee: $60 per passenger for 40 passengers
- Incentive: Fee reduced by $0.50 per passenger for each person over 40
- Revenue: R(x) = (40 + x)(60 - 0.50x) = 2400 + 40x - 0.50x²
- Derivative: R'(x) = 40 - x = 0 □ x = 40 additional passengers
- Optimum: Full bus of 80 passengers at $40/ticket □ Maximum Revenue = $3,200

LECTURER TELEPROMPTER & TEACHING HINTS

Source: MATHalino Maximum Revenue Tour Bus 80 Seats. Demonstrate independent verification of ticket price and passenger revenue.

Applied Optimization: Maximum Revenue

80-Seat Tour Bus: Balancing ticket price vs. passenger demand



Revenue Model Formulation

Base: $\frac{\text{P28,000}}{80} = \text{P350/seat};\ \Delta p = \frac{\text{P1,000}}{80} = \text{P12.50}$

Price: $p(x) = 350 + 12.5x$

Quantity: $q(x) = 80 - 2x$

$$R(x) = (350 + 12.5x)(80 - 2x) = 28{,}000 + 300x - 25x^2$$

$$\frac{dR}{dx} = 300 - 50x = 0 \implies \mathbf{x = 6}$$

Optimal Fare: $p = 350 + 12.5(6) = \mathbf{\text{P425}}$/seat

Passengers: $q = 80 - 2(6) = \mathbf{68}$ seats

Maximum Revenue $= \mathbf{425} \times \mathbf{68} = \mathbf{\text{P28,900}}$

□□ **APPLE PENCIL LIVE WORKSPACE** *[GoodNotes / Notability / Keynote inking area]*

Mathalino Problem 05: Trapezoidal Gutter Capacity

Bending Sheet Metal of Width W into Optimum Cross-Sectional Area

KEY CONCEPTUAL & MATHEMATICAL TAKEAWAYS

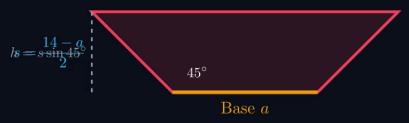
- Sheet of width W divided into 3 equal strips of width w = W / 3
- Sides bent at angle θ to horizontal
- Area: A(θ) = w² (sin θ + sin θ cos θ) = w² sin θ (1 + cos θ)
- Derivative: dA/dθ = w² (2 cos² θ + cos θ - 1) = w² (2 cos θ - 1)(cos θ + 1) = 0
- Optimal angle: cos θ = 1/2 □ θ = 60° (Equilateral trapezoid)

LECTURER TELEPROMPTER & TEACHING HINTS

Source: MATHalino Trapezoidal Gutter Optimization. Highlight the universal geometric beauty: optimal hydraulic radius yields 60° flare.

Engineering Optimization: Trapezoidal Gutter

14-inch metal strip bent at 45°: Maximizing hydraulic carrying capacity



Trapezoidal Cross-Section Calculus

Height: $h = \frac{14-a}{2\sqrt{2}}$

Area: $A(a) = ah + h^2 = \frac{14a - a^2}{2\sqrt{2}} + \frac{(14-a)^2}{8}$

$\frac{dA}{da} = \frac{7-a}{\sqrt{2}} + \frac{a-14}{4} = 0$

$a(2\sqrt{2}-1) = 14(\sqrt{2}-1)$

$a = \frac{14(\sqrt{2}-1)}{2\sqrt{2}-1} \approx \mathbf{3.17}$ inches

Slant Sides: $s = \frac{14-3.17}{2} \approx \mathbf{5.41}$ inches

Concavity: $\frac{d^2A}{da^2} = \frac{1}{4} - \frac{1}{\sqrt{2}} < 0 \implies$ **Maximum**

□□ **APPLE PENCIL LIVE WORKSPACE** *[GoodNotes / Notability / Keynote inking area]*

Active Classroom Challenge & Exit Ticket

Synthesis Problem: Optimize a Cylindrical Storage Silo

KEY CONCEPTUAL & MATHEMATICAL TAKEAWAYS

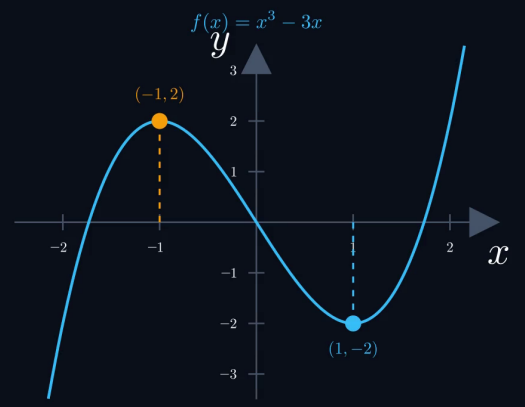
- Challenge: Fixed volume V0 cylinder with hemispherical top
- Minimize total sheet metal surface area S(r, h)
- Identify geometric constraint reducing system to single variable r
- Predict the ratio of height h to radius r for minimum cost

LECTURER TELEPROMPTER & TEACHING HINTS

Exit ticket problem for live classroom participation. Give students 5 minutes with Apple Pencil to set up the objective function.

Unified Master Sequence: The Cubic Manifold

f(x) = x³ - 3x: Continuous Kinematic Derivation



Second Derivative & Concavity Verification

$$f''(x) = 6x$$

$f''(-1) = -6 < 0 \implies$ Concave Down (∩)

Traps LOCAL MAXIMUM at (-1, 2)

$f''(1) = +6 > 0 \implies$ Concave Up (∪)

Holds LOCAL MINIMUM at (1, -2)

APPLE PENCIL LIVE WORKSPACE *[GoodNotes / Notability / Keynote inking area]*

Curriculum References & Citations

Primary Source: MATHalino Differential Calculus Review

KEY CONCEPTUAL & MATHEMATICAL TAKEAWAYS

- Primary Source: MATHalino Engineering Review (mathalino.com)
- 01-04: Number Problems in Maxima and Minima
- 09-11: Rectangular Lot Problems in Maxima and Minima
- 15-17: Box Open at the Top Optimization
- 03: Maximum Revenue Tour Bus (80 Seats)

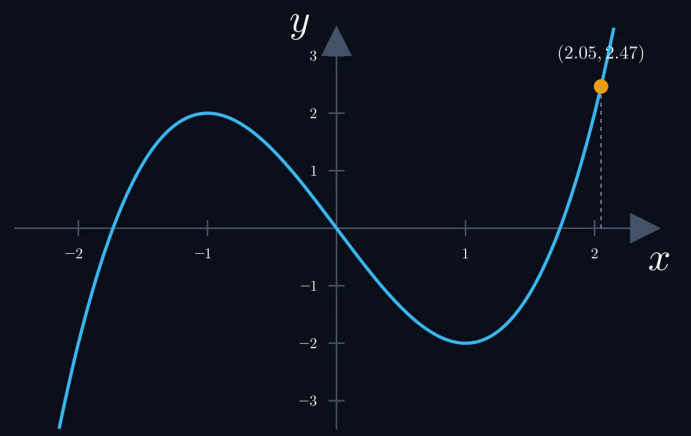
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Formal citations of all MATHalino source pages. Emphasize that all numerical figures and solutions are independently derived.

The Cubic Function Landscape

Tracing vertical displacement and turning behavior along f(x) = x³ - 3x

$f(x) = x^3 - 3x$



APPLE PENCIL LIVE WORKSPACE *[GoodNotes / Notability / Keynote inking area]*